



Detecting Suspicious Activities in Waste Transport Data via Temporal and Network-Aware Change Detection

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Abstract. Waste management networks are essential for environmental protection and public health, but vulnerable to regulatory evasion, fraud, and illegal trading. Detecting potentially illicit activities in the network requires robust anomaly detection systems. However, the complexity of interactions between heterogeneous entities such as recycling companies, individuals, and other organisations combined with temporal irregularities and network dynamics makes conventional fraud detection approaches less effective. In this study, we introduce a dynamic graph-based framework that combines statistical change detection methods, the Page-Hinkley and CUSUM tests, alongside a deep learning model, LSTM-VAE, to detect suspicious activities in the Portuguese waste management network. Using real-world waste transfer records, we engineered temporal and network features to reveal a wide range of anomalies, including abrupt shifts in activity and unusual connectivity patterns, such as those involving collusive triangles. The evaluation was based on four pre-labeled anomalous companies identified by regulators. Our results show that while individual methods excel at detecting certain behaviors, their combination provides robust coverage of diverse anomaly types, with each anomalous company identified by at least three techniques. This approach demonstrates the importance of integrating temporal and network-based analysis, offering regulatory authorities a scalable tool to prioritize inspections, enhancing accountability in the waste management network.

Keywords: fraud detection · change detection · autoencoders · waste management

1 Introduction

Change detection methods are powerful techniques for identifying significant deviations in data over time, such as sudden or gradual shifts in behavior. In

network data, such as records from the waste management network, these methods can detect sudden or gradual changes that point to suspicious behavior, as illicit activities are not usually continuous and may often appear as occasional changes in the operational and network behavior of the entities involved.

Detecting these behaviors is especially important in waste management, because the sector is vulnerable to problems such as regulatory non-compliance, misreporting, illegal trading and waste laundering, which can have significant environmental and public health costs. Whereas traditional fraud detection methods often fail to capture the full temporal and structural complexity of the data, the combination of change detection and deep learning approaches provides a comprehensive framework for detecting suspicious behavior in waste transport networks.

Aggregating data over time often masks short-term fluctuations or irregularities that may indicate suspicious behavior. For instance, a company might show irregular monthly totals of waste received per month as result of an error or some fraudulent transactions occurred in a month, signals that get smoothed out in aggregation. Similarly, traditional tabular or non-network models ignore the structure of interactions between actors. In the waste management context, the relationships between senders and receivers, especially the role of individuals (not companies), can reveal critical patterns such as connections with suspicious nodes or sudden changes in network connectivity that flat-data models cannot capture.

Moreover, actors involved in irregular activities often mimic normal behavior when observed through static metrics or aggregate summaries. By focusing on how behavior evolves over time through change detection techniques, we can better capture deviations from an entity’s own historical patterns. This temporal perspective is particularly important in uncovering events or frauds that are abrupt or timed as in the context of waste management.

In this framework, we explore two change detection techniques, the Page-Hinkley (PH) test and the Cumulative Sum (CUSUM) test, and a deep learning approach based on long short-term memory variational autoencoders (LSTM-VAE). While these methods are well-established in fields like finance, cybersecurity, and manufacturing, their use in environmental and regulatory domains such as waste management remains largely unexplored. Moreover, their integration with dynamic network features for monitoring structural behavior in real-world applications is relatively less explored. This is often due to the limited availability of structural data or the complexity involved in modeling temporal interactions within networks.

This work fills that gap by adapting and evaluating these methods for the waste management network, allowing us to monitor the temporal evolution and behavior of the companies, in order to detect suspicious behaviors and potential fraud within the system. As inputs of these detection frameworks, we introduce a novel use of features representing the temporal, operational and structural behavior of waste operating companies, such as the participation in network triangles, connectivity to individuals participating in triangles, node centralities,

and waste quantities sent and received. By doing so, we are able to capture a broad range of suspicious behaviors, such as sudden spikes or unusual patterns in the temporal, operational and network aspects of the companies.

Our results show that, while the PH test on Connectivity with Individuals in Triangles yielded the best results, it is the combination of several methods that provides the most comprehensive coverage over different types of anomalies. The combination of five different methods ensured that the small set of labeled anomalous companies available for evaluation were always detected by at least three of the techniques. This demonstrates the value of integrating temporal and network-based analysis for effective detection of suspicious activities in waste management networks.

2 Literature Review

Change detection in data streams is an important area of research, with broad applications in fields such as industrial monitoring, finance, and cybersecurity, as well as in adaptive learning systems and real-time analytics. The core goal is to identify points in time where the underlying data distribution changes, which can signal important events, anomalies, or emerging patterns that require attention.

Change detection techniques specifically target the identification of significant shifts in data distributions over time. Among the most widely used are the Cumulative Sum (CUSUM) and Page-Hinkley (PH) tests. The CUSUM test monitors the cumulative sum of deviations from a target value and is effective for detecting small, persistent changes in mean or variance. The Page-Hinkley test, on the other hand, is designed to detect abrupt changes in the average of a data stream and is known for its robustness and low false alarm rates [3]. Both methods have been extensively applied in real-time anomaly detection scenarios, such as industrial process control, financial fraud detection and event detection in social networks [1, 3, 9]. Other notable change detection methods include sliding window approaches, which compare distributions in reference and current windows to detect changes [3]. Non-parametric techniques, such as those based on distance or discrepancy measures between data windows, are also common and can handle both continuous and discrete data without strong distributional assumptions [3]. Martingale-based methods and Bayesian approaches have also been applied, offering flexibility in handling concept drift and providing statistical guarantees [7, 11]. Principal Component Analysis (PCA)-based frameworks are used for multidimensional data streams, capturing changes in the underlying structure of high-dimensional data [10]. These statistical techniques are valued for their simplicity, low computational requirements, and ability to provide statistical guarantees on the reliability of detected changes [8]. However, they may struggle with non-linear or highly complex temporal dependencies, and they are sensitive to parameter choices and noise [1].

In anomaly detection in sequential data and sequential graph data, deep learning-based models, such as Graph Neural Networks (GNN's) [4], transform-

ers [5] and LSTM autoencoders [6] are particularly prominent. However, traditional machine learning models [2, 14] are also frequently used.

Change detection techniques, such as CUSUM and PH tests, complement these methods by monitoring cumulative deviations in data streams, helping to identify distributional shifts that signal potential anomalies, in offline or real-time applications.

Further network-aware change detection exploits structural information from connections, interaction patterns, and higher-order paths between entities, enabling the capture of relational anomalies that are not apparent in isolated temporal analyses. Incorporating such network topology dynamics has been shown to significantly improve the accuracy and robustness of detecting malicious activities in complex, interconnected systems [13].

Overall, the importance of change detection in data or network streams lies in its ability to provide early warnings of system failures, fraud, or other critical events, enabling timely intervention and adaptation [1, 3]. While traditional statistical methods offer interpretability and efficiency, deep learning models provide greater flexibility and performance in complex environments. A combination of these approaches is often beneficial, leveraging the strengths of each to achieve robust and accurate change detection in streaming data.

In the context of waste management, literature on anomaly detection remains sparse. A notable exception is the work of Nirbhaya et al. (2024) [12] who proposed a graph-based framework for analyzing the e-GAR waste transportation data. The authors employ a hybrid approach combining network-based analysis and machine learning to detect anomalies in the waste transportation data. Their method integrates sub-graph detection, community structure analysis, and centrality measures to extract structural features, along with custom edge weight imbalance metrics. Machine learning techniques, including clustering, autoencoders, and density-based methods, are then applied for detecting anomalies. Their approach demonstrated the value of network analysis over traditional transaction-level methods, particularly for detecting edge weight imbalances and unusual network structures. Our work extends this approach by introducing temporal analysis through change detection and deep learning, to detect anomalies in the network and operational behavior over time.

3 Problem Context

Waste transportation involves the systematic movement of waste materials from producers to authorized facilities for treatment, recycling, or disposal. Waste management is an essential sector, as it can have a critical impact on environmental protection and public health. Therefore, waste management is subject to strict regulations to ensure its adequate functioning and minimal environmental and social costs.

Waste management regulations often impose financial and administrative burdens, and some entities may engage in falsifying, misreporting, or omitting waste transports to avoid regulatory obligations, reduce costs, and evade taxes or

fees associated with proper waste disposal. Transporters may misreport the type of waste being moved to bypass restrictions on hazardous materials or to avoid higher disposal costs. Additionally, organized criminal schemes can arise: some types of waste, such as scrap metals and car parts, are significantly valuable and can be easily taken from construction sites, public infrastructures or vehicles. Since these types of waste can be sold to recycling facilities, they become attractive targets for theft and illegal trading. Other entities may engage in waste trafficking operations, which take advantage of regulatory inconsistencies across countries to turn a profit, which can have a severe environmental and public health impact on the destination countries.

Therefore, the regulatory authorities in Portugal are interested in detecting illegal and criminal practices and ensuring transparency and regulatory compliance in the Portuguese waste management network, and they run periodic inspections on the companies involved. However, monitoring and analyzing large-scale waste transport records to identify suspicious patterns or anomalies is a complex task. Therefore, a systematic, accurate and efficient approach is needed.

For the purpose of developing such an approach, we have gained access to waste transportation records. The e-GAR is an electronic document that is required to accompany each instance of waste transport within the national territory. This document is issued by any of the entities involved in the transfer, and subsequently validated by the waste recipient, and includes comprehensive information such as the identification of the senders and receivers, the timestamps of the operations, the quantities, the types of waste materials transferred between entities, and the types of operation.

3.1 Data Set

The data set used in this study comprises a large collection of anonymized e-GAR records covering waste transportation activities carried out over a year in Portugal. Following thorough data cleaning and preprocessing, the final data set contains more than 3.5 million transfer records, comprising over 111,000 unique senders and 1,591 unique waste receivers. Each transport record encapsulates rich contextual information that supports advanced data-driven analysis, including anomaly detection and network-based modeling aimed at identifying irregular or suspicious patterns in waste movement.

Although the data set is primarily unlabeled, a very limited subset of information was given by the data provider identifying four users confirmed to be involved in suspicious or fraudulent activity. These labeled nodes serve a critical role in evaluating the performance of the models applied in this work. Given the severe class imbalance and scarcity of known anomalies, these few labeled instances are used exclusively for evaluation purposes, rather than for supervised training, to maintain the integrity of an unsupervised or weakly supervised anomaly detection framework.

4 Methodology

4.1 Data Preprocessing

Before conducting our analysis, we performed several data preprocessing steps to ensure the quality of the data. Firstly, we removed all e-GAR documents that had invalid states, such as ‘Nullified’, ‘Rejected’, or ‘Deleted’. Additionally, we removed residual transfers from 2021 and 2023, restricting our analysis to waste transfers taking place in 2022. Operations related to civil construction, identified by the code ‘OBRAS RCD’, were also excluded from the dataset, based on guidance from inspectors who indicated that these records were not relevant for the objectives of our study.

For the temporal aspects of our analysis, there were several time references available in the data: the emission date of the document, the start of the transport, and the end of the transport. We chose to use the date corresponding to the beginning of the transport, as this most accurately reflects the actual activity of the waste operators.

As for the quantity aspect of the data, there were two features available: ‘quantity’, filled by the entity emitting the document, and ‘quantity received’, filled by the company receiving the waste. We observed many inconsistencies between these two features, which could be due to weighing discrepancies or data entry errors. Since the ‘quantity received’ is generally filled in by waste operator companies that have more experience and accuracy in this process, we used this value as our main reference for waste quantity. However, in rare cases where ‘quantity received’ was higher than 50,000 kg and the ‘quantity’ field wasn’t, we used the value from ‘quantity’ instead, as such high values are exceptional and likely to be errors if not supported by both features.

After these cleaning steps, the resulting dataset contained over 3.5 million transfer records, with more than 111,000 unique senders and 1,591 unique waste receivers. Using this cleaned data, we constructed a network representing the waste transfers between the entities. The nodes in the network correspond to the senders and receivers of waste, which can be individuals, waste recycling companies or other organizations. The directed edges in the network represent the flow of waste from sender to receiver, and are attributed with the quantity of waste, the date of transfer, and the type of waste. Nodes are further attributed with the type of entity, distinguishing most importantly between individuals and companies. This network structure provided the foundation for our analysis.

4.2 Feature Creation

To perform anomaly detection in the companies of the waste management network, we constructed a set of time series features derived from one year of e-GAR records. These features were designed to capture both the structural characteristics of the network, using Social Network Analysis (SNA) metrics, and the activity patterns of each company, with the goal of revealing unusual dynamics indicative of illicit activities. The features were calculated with daily, weekly,

and monthly granularity, to allow for testing of the temporal analysis at different scales. Below, we present and justify each of the constructed features.

Number of Individual Senders: This feature represents the count of unique individuals who sent waste to each company in a given time period. While companies can receive waste from other companies, individuals, and other types of organizations such as public institutions, individual senders are frequently involved in irregular or illegal transactions, such as the sale of valuable waste that has been stolen or improperly disposed. A sudden increase in the number of individual senders may point to potential involvement in these illicit transactions.

Waste Quantity Received: This feature consists of the total amount of waste, in kilograms, received by each company during a given time period. Monitoring the temporal evolution of this quantity helps identify unusual spikes or drops in a company’s activity that could signal suspicious behavior.

Degree Centrality: This feature consists of the number of unique direct connections (inbound or outbound) that a company has with other entities in the network during a given time period. It represents the company’s level of interaction and influence in the system. With this feature, we are interested in finding unusual changes that suggest suspicious behavior.

First Principal Component (PCA) of Activity Features: To combine the multivariate information from the three features presented above (number of individual senders, waste quantity received, and degree centrality) we applied Incremental Principal Component Analysis (PCA) on the standardized versions of these features. The first principal component, which captures the dominant direction of variance, is then used as a new feature. This helps to reduce dimensionality and highlights composite activity patterns that may not be evident from any single feature.

Connectivity with individuals in triangles: This feature consists of the sum of the number of triangles that the individual waste senders connected to a company participate in. It measures a company’s indirect exposure to suspicious behavior in the network, based on the assumption that individual senders involved in many triangles are more likely to be participating in illicit activities. This feature thus highlights companies that may be receiving stolen waste from individuals or cooperating in illicit schemes with them.

Waste Quantity Balance: This feature is computed as the difference between the total waste quantity sent and the sum of the waste quantity received and the initial stock at the start of each period¹:

$$Balance = QuantitySent - QuantityReceived - InitialStock.$$

¹ The initial stock of the year was obtained through an additional dataset, MIRR, which contains annual reports of waste transported and stored from eligible organisations, and was then recomputed for each following period using the Balance.

This feature can highlight companies whose reported outflows do not align with their reported inflows, and particularly, high positive imbalances of companies that are not merely waste producers can indicate underreporting of received waste.

4.3 Detecting Changes in Dynamic Graph Data

To detect suspicious activities within the waste management network, we employed two complementary statistical change detection techniques: the Page-Hinkley test and the Cumulative Sum (CUSUM) test. These methods were applied to the time series features described in the previous section, with the underlying assumption that fraudulent or illicit practices such as misreporting waste quantities or concealing illegal waste transfers are unlikely to occur continuously. Therefore, they are likely to manifest as occasional deviations from normal activity patterns, which we aim to identify by applying these change detection techniques.

The Page-Hinkley test is a sequential analysis technique designed to detect sudden changes in the mean of a time series. Formally, for a sequence of observations $x_1, x_2, \dots, x_t$ the test computes the cumulative sum of adjusted deviations from the running mean, defined as:

$$PH_t = \sum_{i=1}^t (x_i - \mu - \delta) \quad (1)$$

where μ is the estimated mean of the process up to time t , and δ is the parameter that controls the test's sensitivity to small changes. A change is detected when the difference between the current cumulative sum and its historical minimum (or maximum, depending on the direction of change) exceeds a predefined threshold. In our implementation, both the threshold and the delta parameter are scaled by each company's historical standard deviation (σ), (threshold = $x\sigma$, $\delta = y\sigma$), to account for differences in scale across companies. The strength of each detected change is quantified as the difference between the cumulative sum at the time of detection and the minimum (or maximum) cumulative sum registered before this change:

$$\text{Change Strength}_{PH}(t) = PH_t - \min_{1 \leq j \leq t} PH_j \quad (2)$$

To allow for comparison between companies, it is further scaled by each company's standard deviation. At each change detected, the cumulative sum is reset. This test was implemented using the river library in Python.

The Cumulative Sum (CUSUM) test was employed to identify gradual and sustained changes in the mean of the time series. The CUSUM test maintains two separate cumulative sums to detect both positive and negative changes. For each time point t , the positive and negative cumulative sums are updated as follows:

$$S_t^+ = \max(0, S_{t-1}^+ + x_t - \mu - k) \quad (3)$$

$$S_t^- = \max(0, S_{t-1}^- + \mu - x_t - k) \quad (4)$$

where μ is the expected value (target mean) and k is the drift parameter that determines the allowance for minor fluctuations. A change is detected when one of the cumulative sums exceeds the specified threshold, which, similarly to the Page-Hinkley test, is scaled by the company's standard deviation for comparability. The strength of each detected change is calculated as the cumulative sum of deviations from the target mean that exceed the drift parameter, up to the point of detection:

$$\text{Change Strength}_{\text{CUSUM}}(t) = \max(S_t^+, S_t^-) \quad (5)$$

It is further scaled by the company's standard deviation. By employing these two complementary techniques, we capture both abrupt and gradual changes in the behavior of the companies within the waste management network.

4.4 LSTM-VAE

The Long Short-Term Memory Variational Autoencoder (LSTM-VAE) is a deep generative model that combines the ability of LSTM networks to model sequential data with the probabilistic latent variable framework of Variational Autoencoders. While the variational autoencoder learns a compressed, probabilistic representation of normal activity that is robust to noise and variability in the data, the LSTM component is able to capture long-term dependencies and temporal dynamics in the companies' behavior, which is critical for distinguishing between normal fluctuations and anomalies.

In our implementation, the LSTM-VAE was trained on 4 weekly time series features for each company: weekly waste quantity balance, weekly number of individual senders, weekly degree centrality, and weekly number of triangles. These features were selected to obtain a broad representation of the company's activity patterns, particularly in aspects where suspicious behavior could manifest, such as sudden spikes in waste quantities or unusual changes in network connectivity.

Prior to training, each feature was scaled using a standard scaler fitted on non-zero values, except for the number of triangles, where zeros were included in the scaling process. This design choice is due to the fact that most features have a high percentage of zeros, because of the periodicity of many companies' activity, and their inclusion could bias the model towards treating non-zero activity as anomalous. On the other hand, for the number of triangles, zero is the expected (normal) value and significant non-zero values are potentially suspicious.

The LSTM-VAE was implemented using the Keras library in Python. The model consists of an encoder LSTM that processes input windows of length t . Each input has shape (t, n) , where t is the window size and n is the number of features. The encoder outputs two vectors, the mean and the log-variance,

which define the approximate posterior distribution over the latent variables. A latent vector of dimension d , is then sampled from this distribution using the reparameterization trick. This sampled latent vector is passed to a decoder LSTM, which reconstructs the original input sequence. The model is trained to minimize a loss that combines two terms: the reconstruction loss (measured as the mean squared error between the original input and its reconstruction), and the Kullback–Leibler (KL) divergence between the approximate posterior and a standard normal prior.

For model training, we constructed sliding windows over each company’s feature time series. The companies that have been flagged as anomalous were excluded from the training, in order to learn the normal activity pattern. After training, anomaly detection was performed by computing the reconstruction error for each window:

$$\text{Reconstruction Error} = \frac{1}{T \cdot N} \sum_{i=1}^T \sum_{j=1}^N (x_{i,j} - \hat{x}_{i,j})^2$$

High reconstruction errors indicate that the observed sequence deviates from the patterns seen during training and is therefore flagged as potentially anomalous.

5 Results Discussion

5.1 Anomaly Scoring

To combine the results from the change detection methods and the LSTM-VAE model, we used a ranking-based system to identify suspicious companies.

For the Page-Hinkley (PH) and CUSUM tests, we calculated the total change strength for each company, as the sum of the strengths of all detected anomalies over time, as defined in the Sect. 4.2. Companies were then ranked on this total, with higher values indicating stronger evidence of anomalous behavior. For the LSTM-VAE, companies were ranked by their mean reconstruction error, where higher errors suggest the model struggled to replicate their activity patterns, signaling potential anomalies. For easier interpretability and comparison, we converted the ranks into standardized scores, based on how many companies were subject to these methodologies:

$$\text{Score} = \frac{\text{Rank} - 1}{\text{Number of Companies} - 1} \quad (6)$$

This score results in values between 0 (least anomalous) and 1 (most anomalous). We utilized five methods to obtain anomaly scores for each company:

- CUSUM test on the First Principal Component of PCA on the Activity Features (CUSUM-PCA);
- PH test on the First Principal Component of PCA on the Activity Features (PH-PCA),

- CUSUM test on the Connectivity with individuals in triangles (CUSUM-Triangles),
- PH test on the Connectivity with individuals in triangles (PH-Triangles),
- LSTM-VAE.

These methods were selected because they yielded the best results according to our evaluation (which will be discussed in the Sect. 5.2) and because they offer complementary insights into anomalous behaviour, balancing sensitivity to sudden shifts, gradual drifts, and complex behavioral changes. The rank-based scoring system was chosen due to its robustness to the skewed distributions of the change strengths and mean reconstruction errors, ensuring also a more consistent comparison of the results from these different methods.

Finally, the results of the different methods were combined into an overall anomaly score, consisting of the median of the five methods. The median was chosen as the aggregation measure due to the complementarity of the approaches:

5.2 Evaluation

To assess the effectiveness of our anomaly detection framework, we evaluated the performance of the individual methods on the set of four pre-labeled anomalous companies. These companies were identified by regulatory authorities as having engaged in illicit practices, providing a ground-truth benchmark for validation. The anomaly scores for each company obtained by the five methods, and the overall score, are summarized in Fig. 1.

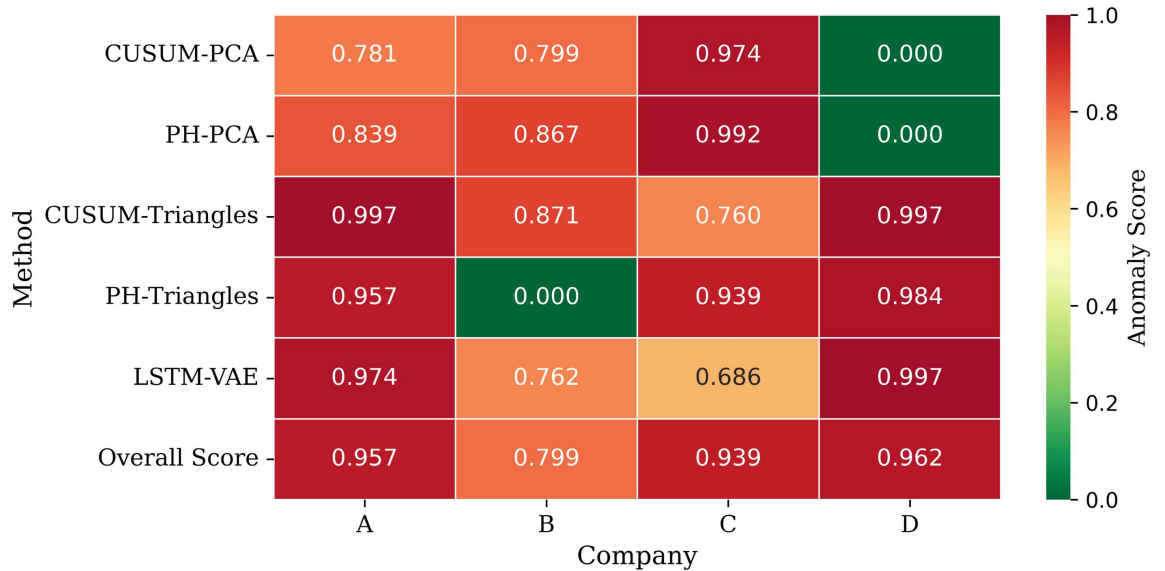


Fig. 1. Anomaly Scores obtained in each method by the pre-labeled anomalous companies.

The CUSUM test on Connectivity with Individuals in Triangles (CUSUM-Triangles) demonstrated the strongest performance, as it was able to detect

all anomalous companies with high anomaly scores, particularly companies A and D. This aligns with expectations, as illicit activities often involve covert collaborations between entities, which manifest as unusual connectivity patterns.

Notably, all methods had a strong performance in at least three of the companies. The CUSUM-PCA, PH-PCA and PH-Triangles failed to detect one of the companies each. For instance, company D was undetected by the PCA-based change detection methods, but was identified with a strong anomaly score for the remaining 3 methods. This highlights the complementarity of the approaches, which cover different aspects of anomalous activity in the waste management network. The combination of the five methods through the median thus resulted in a high anomaly score (≥ 0.8) for all four anomalous companies.

Despite the severely limited validation dataset, these results suggest that performing a temporal analysis on structural and network-based signals, provides a useful framework for flagging anomalous activity in the waste management network, for the purpose of regulatory inspections.

5.3 Analysis of the Top Anomalous Companies

We now analyse three of the top scoring companies, including the previously presented company A for comparison, as it obtained high scores across all different methods and also a high overall score. The company with the highest anomaly score (0.983) is presented in Fig. 2.

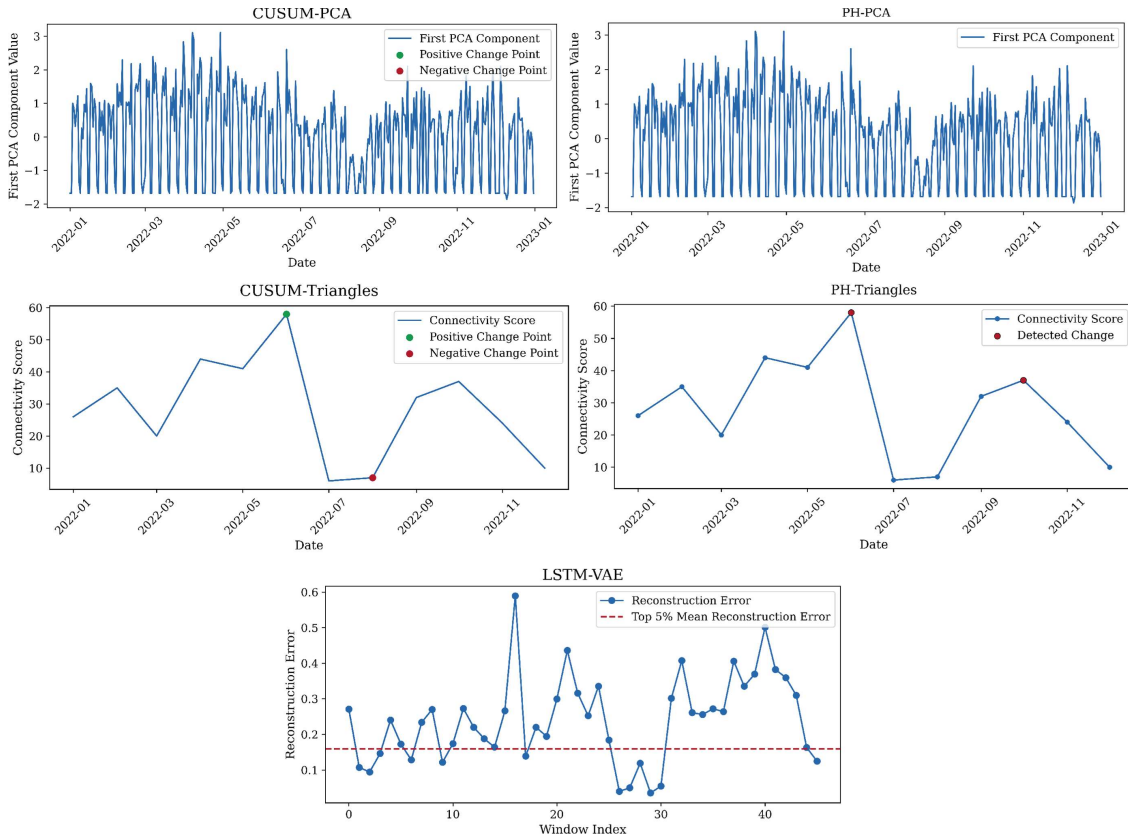


Fig. 2. Results of the highest scoring company across the five methods.

This company was identified by both the triangle-based methods and the LSTM-VAE, while no significant changes were detected in the first component of PCA. The high scores from the CUSUM and PH tests (0.996 and 0.99) on connectivity with individuals in triangles indicate that this company’s connections with individuals involved in triangles oscillated between periods of high and low connectivity. The LSTM-VAE also assigned a high anomaly score (0.983), as the company showed extremely high reconstruction errors in certain periods, alternating with periods of more typical behavior. This suggests that the company’s activity was irregular, with periods of highly unusual patterns.

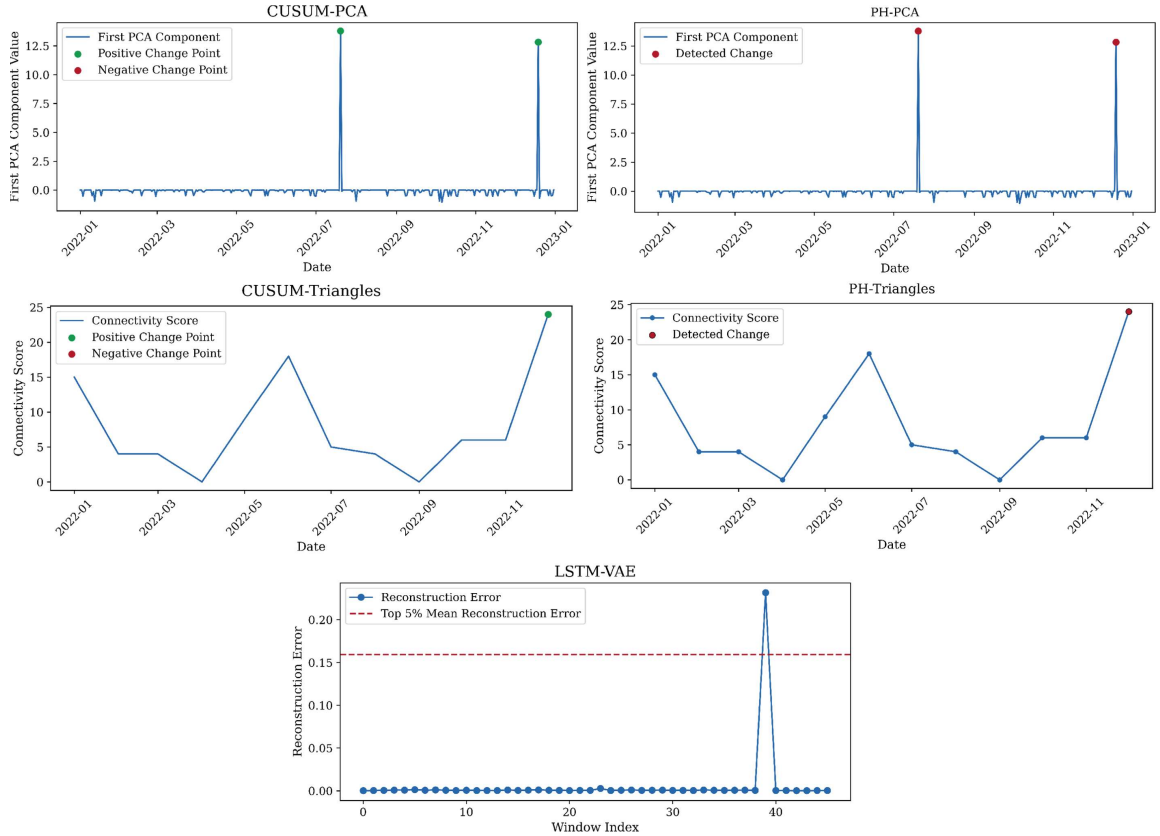


Fig. 3. Results of another of the highest scoring company across the five methods.

The company presented in Fig. 3, with an overall score of 0.965, was detected by all five methods. The PH and CUSUM tests (scores 0.975 and 0.996) identified two sharp spikes in the first component of PCA based on activity features, pointing to sudden and significant changes in operational behavior. On the connectivity to individuals in triangles, both the CUSUM and PH tests (scores 0.77 and 0.965) again revealed an oscillation between high and low connectivity, though less pronounced than in the previous case. The LSTM-VAE assigned a moderate score (0.554), as it only detected a high reconstruction error in one period, indicating that, according to this method, most of the company’s temporal behavior was closer to the normal patterns learned by the model.

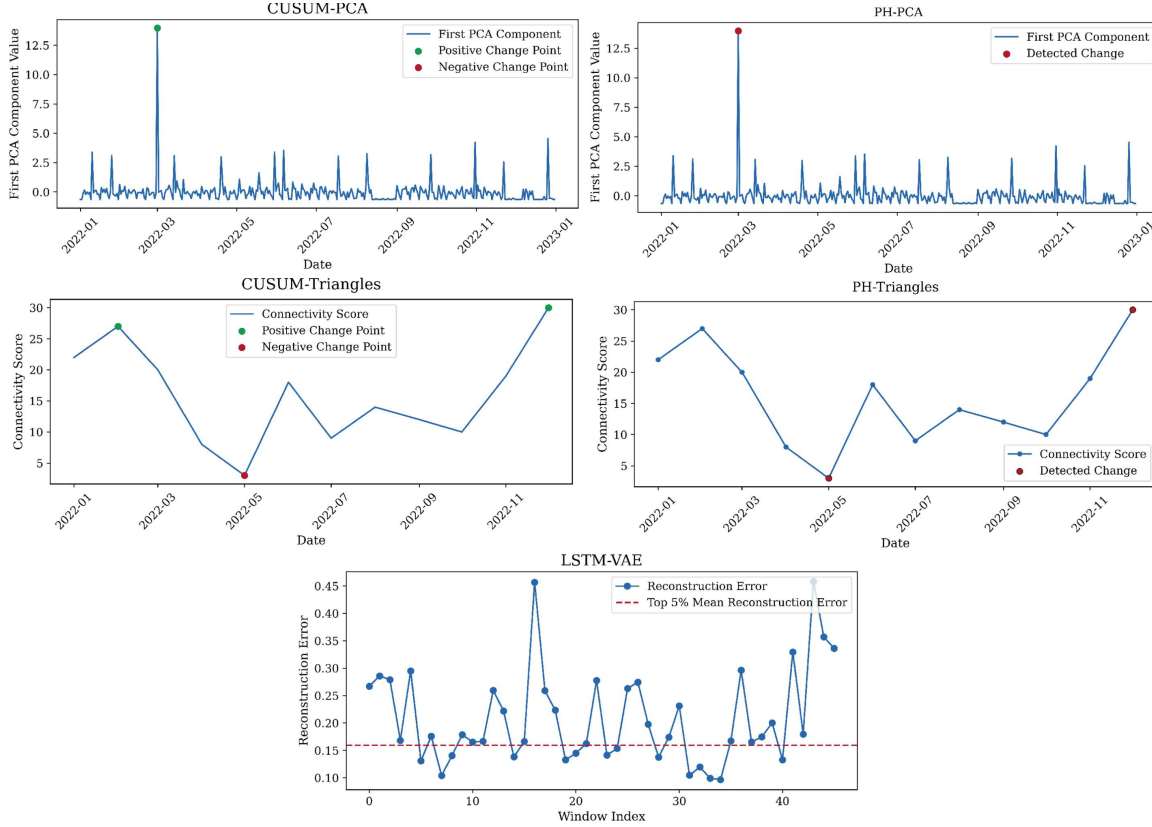


Fig. 4. Results of the pre-labeled anomalous company A across the five methods.

Company A, shown in Fig. 4, with an overall score of 0.957, was also identified as anomalous by all methods. The PH and CUSUM tests (scores 0.781 and 0.839) detected a steep spike in the first component of PCA, and the triangle-based tests (CUSUM: 0.997, PH: 0.957) again showed oscillating connectivity patterns with suspicious individuals. The LSTM-VAE assigned a high anomaly score (0.974), as the company frequently exhibited high reconstruction errors, with two periods showing particularly severe deviations from normal behavior. These examples highlight several important points. First, the different methods complement each other, with some detecting anomalies in operational features and others in network connectivity or temporal patterns. Second, the oscillating patterns of connectivity and reconstruction errors suggest that suspicious activity is not constant but occurs in bursts, possibly as a strategy to avoid detection. Finally, the integration of temporal, operational, and structural analysis is essential for capturing the full range of anomalous behaviors, providing regulatory authorities with more reliable and actionable signals for further investigation.

6 Conclusions

In this work, we developed and evaluated a framework for detecting suspicious and potentially illicit activities in the Portuguese waste management network

using dynamic graph analysis. By combining statistical change detection methods, namely the Page-Hinkley and CUSUM tests, with a deep learning approach based on an LSTM-VAE model, we aimed to detect abrupt or gradual changes the activity of the companies, as well as complex temporal patterns that may indicate fraud or non-compliance.

Our methodology relied on features that reflect both the operational and the network behavior of each company within the waste management network. The evaluation of our approach on a small set of known anomalous companies showed that the ensemble of different techniques provided a comprehensive coverage of different types of anomalies.

Overall, our results demonstrate that, considering fraud or illicit activities are not practiced continuously but rather sporadically, performing a temporal analysis can successfully detect irregularities suggestive of suspicious activity in waste management systems. This approach not only supports regulatory authorities in prioritizing investigations but also contributes to greater accountability in environmental protection efforts.

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